

Master Thesis Proposal

Title: Design and Evaluation of an LLM-based Retrieval-Augmented-Generation Prototype for Security Analyst Support in Cyber Threat Intelligence

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Problem Background

Cyber Threat Intelligence (CTI) plays a central role in modern cybersecurity operations by helping analysts proactively identify emerging vulnerabilities, attack techniques, and adversary behavior. However, the constantly changing and dynamic nature of threat information, coming from structured sources such as CVE, CWE, and MISP as well as from unstructured threat reports and security blogs, makes it difficult for analysts to maintain situational awareness and respond quickly to new exploits (Paul et al., 2025; Fayyazi et al., 2024). This challenge is particularly evident in threat hunting, where analysts must continuously correlate new indicators and tactics across diverse and rapidly evolving data streams to uncover hidden adversary activity.

Large Language Models (LLMs) show strong potential for processing and summarizing such data, but their static nature remains a fundamental limitation. Once deployed, an LLM's knowledge base quickly becomes outdated and cannot easily incorporate newly disclosed vulnerabilities without costly retraining (Chaganti, 2024; Paul et al., 2025). In the context of threat intelligence, this lack of up-to-date knowledge can lead to false or incomplete assessments, reducing the reliability and usefulness of analytical outputs when relying on an LLM.

Retrieval-Augmented Generation (RAG) has recently emerged as a promising approach to mitigate this issue by linking LLMs with continuously updated external knowledge sources (Lewis et al., 2020; Gao et al., 2024). However, despite its potential, there is limited empirical understanding of how RAG architectures perform in CTI and threat-hunting contexts and how they can be effectively integrated into analysts' workflows. This gap highlights the need for a structured empirical evaluation and the development of practical design principles that demonstrate how RAG can be applied to dynamic, intelligence-driven cybersecurity tasks.

Task Definition

The objective of this thesis is to design, implement, and evaluate a Retrieval-Augmented Generation (RAG) prototype to support analysts in retrieving and contextualizing Cyber Threat Intelligence (CTI) information during Threat-Hunting workflows. The work focuses on improving the retrieval, contextualization, and preparation of CTI information by combining heterogeneous data sources within a unified, queryable framework enhanced by generative AI.

The primary research focus lies on the analysis and reasoning capabilities of an RAG-augmented LLM, rather than on large-scale data collection. CTI data from available sources (e.g., CVE/NVD, MISP, CISA advisories, and selected vendor reports) will serve as representative inputs to evaluate how retrieval augmentation affects information completeness, factuality, and explainability in analyst workflows.

The target group of this work includes security analysts and threat-hunters who need fast access to accurate, source-grounded intelligence, and security managers overseeing the adoption of AI-assisted tools within Security Operations Center (SOC) workflows.

The expected tasks and deliverables of this thesis are:

1. Prototype Development

- Implement a proof-of-concept system that enables natural-language querying and retrieval-augmented reasoning over CTI data
- Core components include
 - Hybrid retrieval pipeline (BM25 + vector-based search)
 - Context assembly and grounding for LLM-based answer generation
 - Transparent citation mechanism for source traceability
- Optionally, the prototype may be extended with a lightweight agent-based orchestration layer (e.g., via Model Context Protocol, MCP) to coordinate retrieval, analysis, and response generation steps in a modular manner

2. Quantitative and Qualitative Evaluation

- Conduct a quantitative evaluation of retrieval and generation quality using established metrics as described in the relevant literature and the *RAGAS* framework (indicators such as Faithfulness, Context Precision, Context Recall)
- Complement the quantitative evaluation with a qualitative analysis of system outputs, focusing on interpretability, structure, and practical usefulness in representative CTI task scenarios.

3. Reference Model Development

- Derive a conceptual reference model based on empirical results, describing how RAG-LLM systems can be effectively integrated into CTI and threat-hunting workflows
- The model consolidates architectural components, data flows, and interaction patterns observed during implementation and evaluation, focusing on feasibility and practical applicability

Expected outcome:

An empirically grounded and expert-validated reference model for applying RAG-based LLMs in Cyber Threat Intelligence and Threat-Hunting. The findings aim to provide both practical insights for improving analyst workflows and a conceptual foundation for future research on AI-assisted CTI systems.

Research Questions

Main Research Question (MRQ)

How can a Retrieval-Augmented Generation (RAG) system leveraging Large Language Models improve the quality, reliability, and operational usefulness of Cyber Threat Intelligence (CTI) analysis and Threat-Hunting workflows?

Sub Research Questions (SRQs)

SRQ1 – Information Retrieval and Answer Quality

To what extent does retrieval augmentation improve the completeness, factual accuracy, and recency of CTI information generated by an LLM compared to a non-retrieval baseline?

Measurement: A quantitative evaluation based on established Information Retrieval (IR) metrics and RAG-specific evaluation frameworks will be conducted to assess retrieval quality, contextual relevance, and factual grounding of generated CTI outputs. Results will be compared against a baseline LLM without retrieval augmentation to quantify improvements in information coverage and reliability.

SRQ2 – Practical Utility and Workflow Support

Which types of CTI analysis tasks can be effectively supported by a RAG-based LLM system, and how well do the generated outputs align with the informational needs of threat-hunting workflows?

Measurement: The system will be evaluated through scenario-based analysis using representative CTI tasks (e.g., vulnerability assessment, IOC enrichment, threat actor context building). Outputs will be analyzed qualitatively with respect to task coverage, structural clarity, and transparency.

SRQ3 – Conceptual Design:

How can empirical insights from the RAG-LLM prototype be translated into actionable design principles for integrating generative AI into CTI and Threat-Hunting workflows?

Measurement: A structured synthesis of the evaluation results will be conducted to derive a set of design principles and integration guidelines. These principles will be grounded in observed system behavior, architectural decisions, and task-level performance across the evaluated CTI scenarios.

Methods

The methodological framework of this thesis should consist of six phases: Literature Review, Process Analysis, Prototype Development, Evaluation and Reference Modeling.

Literature Review

To establish a solid theoretical and technical foundation for the prototype by analyzing relevant research on Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), and Cyber Threat Intelligence (CTI).

Approach:

At least 20 scientific and technical relevant sources are systematically reviewed to identify:

- Current CTI practices / SOC workflow
- Current RAG architectures and evaluation methods (e.g., Information Retrieval metrics)
- The role and limitations of LLMs in cybersecurity / CTI context
- Best practices for explainable AI and retrieval quality

Expected outcome:

A consolidated overview of state-of-the-art methods and existing gaps, forming the basis for prototype design and evaluation.

Process Analysis / Use Case Definition

To identify relevant CTI workflows and use cases where retrieval-augmented reasoning can provide measurable improvement.

Approach:

Analysis of common SOC/CTI processes based on practitioner documentation and existing frameworks (e.g., MISP workflows, ATT&CK mapping).

Typical analyst tasks to be modeled may include:

- Indicator of Compromise (IOC) enrichment
- Vulnerability summarization (CVE/CWE)
- Adversary behavior mapping (TTP correlation)

Expected outcome:

A representative set of CTI tasks and evaluation queries forming the experimental context for prototype testing.

Prototype Development

To design and implement a functional proof-of-concept system that demonstrates how Retrieval-Augmented Generation can improve the quality, explainability, and recency of CTI analysis.

The prototype should show:

- That CTI data from multiple sources can be efficiently aggregated, contextualized, and queried in natural language, enabling faster and more transparent analysis.

- That retrieval augmentation increases the *completeness and factual accuracy* of generated intelligence compared to an LLM alone.
- The prototype may also illustrate how contextualized and retrieved CTI information can be synthesized into concise, source-grounded recommendations that support analysts during threat-hunting tasks.

Approach:

The prototype will consist of three main components:

1. Data ingestion and indexing: Aggregation of heterogeneous CTI data (CVE/NVD, CISA advisories, MISP, vendor blogs) and conversion into retrievable document chunks (JSON/CSV)
2. Hybrid retrieval layer: Combined lexical (BM25) and vector-based search with optional reranking for result prioritization
3. RAG orchestration: Integration with an open-source LLM (e.g., Llama 3) for context-based answer generation, including automatic citation of retrieved sources

The system will be implemented using Python with frameworks such as *LangChain* and *ChromaDB* for retrieval indexing. The focus lies on using **open-source** technology only.

Optional: Agent-based Extension (Model Context Protocol)

As an optional extension, the RAG pipeline may be exposed as a callable service within a lightweight agent-based orchestration using the Model Context Protocol (MCP). In this setup, specialized agents can coordinate multi-step CTI analysis tasks by invoking the RAG component as a tool (e.g., for retrieval and analysis). This extension is exploratory in nature and does not form part the core prototype evaluation.

Expected outcome:

A fully functional prototype capable of retrieving relevant CTI context, generating transparent, source-grounded summaries, and providing measurable output for performance evaluation.

Evaluation

To empirically evaluate the performance and utility of the developed prototype. The evaluation aims to determine to what extent retrieval augmentation improves factual accuracy, completeness, and usefulness of generated CTI outputs.

Approach:

- Quantitative Evaluation:
 - Use of Information Retrieval metrics (IR) to assess document relevance and ranking quality
 - Use of RAG-oriented evaluation frameworks to measure factual grounding, contextual relevance, and answer quality of generated outputs
 - Benchmark comparison against an LLM baseline without retrieval augmentation using the same query set
 - optionally: Execution of ablation studies (BM25-only vs. vector-only vs. hybrid) to identify the most effective retrieval strategy
- Qualitative Evaluation:
 - Scenario-based analysis using representative CTI tasks (e.g., vulnerability analysis, IOC enrichment, threat actor context building)
 - System outputs will be assessed with respect to task coverage, contextual completeness, transparency, and interpretability

Expected outcome:

Quantitative evidence that retrieval augmentation improves CTI information quality and qualitative confirmation that the system supports analysts in interpreting and contextualizing intelligence faster and more transparently.

Reference Modeling

To consolidate and generalize the empirical findings from the prototype evaluation into a structured concept that can guide the practical integration of RAG-based LLM systems into Cyber Threat Intelligence (CTI) and Security Operations Center (SOC) workflows.

Approach:

Create a conceptual integration model that visualizes the technical and procedural interaction between CTI data sources, the retrieval pipeline, and the analyst interface.

This includes:

- Data flow from ingestion to retrieval and answer generation
- Modeling of analyst–system interactions, specifying where analysts input queries, inspect retrieved context, and verify or refine the generated threat intelligence
- Information governance considerations (traceability, update frequency, auditability)

And document technical and organizational enablers such as:

- Data quality requirements
- Infrastructure considerations
- Maintenance processes for continuous data ingestion and model updates

Expected outcome:

A concise, transferable reference concept that outlines how RAG-based generative AI systems can be systematically implemented in CTI and SOC environments. The model will combine architectural blueprints, process integration steps, and governance recommendations, serving as a practical implementation guide for organizations aiming to adopt retrieval-augmented AI for cyber threat analysis and threat hunting.

Expected Type of Results per Method

Overview: Research Questions – Methods – Results

<i>Research Questions / Hypotheses</i>	<i>Methods Applied per Research Question</i>	<i>Expected Type of Result per Method</i>
MRQ: <i>How can a Retrieval-Augmented Generation (RAG) system leveraging Large Language Models improve the quality, reliability, and operational usefulness of Cyber Threat Intelligence (CTI) analysis and Threat-Hunting workflows?</i>	Literature Review, Process Analysis, Prototype Development, Evaluation	Functional prototype showing measurable improvement in CTI information quality and contextualization compared to a non-retrieval baseline
SRQ1: <i>To what extent does retrieval augmentation improve the completeness, factual accuracy, and recency of CTI information generated by an LLM compared to a non-retrieval baseline?</i>	Quantitative Evaluation (IR & RAGAS metrics), Prototype Experimentation	Quantitative results demonstrating improved retrieval quality, contextual relevance, and factual grounding compared to a non-retrieval baseline
SRQ2: <i>Which types of CTI analysis tasks can be effectively supported by a RAG-based LLM system, and how well do the generated outputs align with the informational needs of threat-hunting workflows?</i>	Scenario-based System Evaluation, Comparative Analysis against a non-retrieval baseline	Task-oriented evaluation results showing improved contextual coverage, clarity, and transparency of CTI outputs in representative analysis scenarios
SRQ3: <i>How can empirical insights from the RAG-LLM prototype be translated into actionable design principles for integrating generative AI into CTI and threat-hunting workflows?</i>	Reference Modeling, Synthesis of Evaluation Results	Validated integration concept outlining design principles and organizational enablers for SOC adoption

Timeline

Milestone Plan / Work Steps

M#...Milestone

P#...Private planning goal

Nr	Name	completion date
M1	Submission of revised thesis proposal draft	08 October 2025
M2	Submission of "Masterarbeit Thema" form (with company supervisor signature)	23 November 2025
P1	Literature review completed (foundation on CTI, LLMs, and Retrieval-Augmented Generation)	Mid-December 2025
P2	Research design finalized (definition of CTI tasks, data sources, and evaluation framework)	End of December 2025
M3	Submission of final proposal	18 January 2026
P3	CTI data ingestion and normalization implemented; database schema	End of January 2026
P4	Hybrid retrieval pipeline developed (BM25 and vector retrieval, reranking integrated)	Mid-February 2026
P5	RAG orchestration layer implemented (context retrieval, prompt construction, and LLM response generation)	End of February 2026
P6	Prototype testing - Phase I: internal validation of retrieval and generation quality	Mid-March 2026
P7	Prototype testing - Phase II: analyst-oriented demonstration and threat-hunting scenario evaluation	End of March 2026
P8	Reference modeling and consolidation of evaluation results	Early April 2026
M4	Submission of first full thesis draft (<i>Erstversion</i>) for supervisor feedback	07 April 2026
P9	Incorporation of feedback and refinement of prototype, evaluation, and thesis	April 2026
M5	Submission of final thesis (digital and printed)	10 May 2026

Proposed Structure of the Thesis

**The following structure represents the current planned outline and expected page count of the thesis. Minor adjustments may be made during the writing process to reflect the final scope and results of the research.*

Declaration of authenticity (1 page)

Kurzfassung ($\frac{3}{4}$ page)

Abstract ($\frac{3}{4}$ page)

Acknowledgements ($\frac{1}{4}$ page)

Table of contents (1 page)

1. Introduction

1.1 Problem statement and motivation (1 page)

1.2 State of the art ($\frac{3}{4}$ to 1 page)

1.3 Research questions and objectives ($\frac{1}{2}$ page)

2. Methodology (1 page – overview research methodology as a figure)

2.1 Literature review (1 to $1\frac{1}{2}$ pages)

2.2 Prototype development

2.2.1 System overview (1 to $1\frac{1}{2}$ pages)

2.2.2 Data preparation and preprocessing ($1\frac{1}{2}$ pages)

2.2.3 Embedding and retrieval strategy ($\frac{1}{2}$ to $\frac{3}{4}$ page)

2.2.4 Answer generation and citation ($\frac{1}{2}$ page)

2.2.5 Agent based extension (MCP) (optional)

2.3 Evaluation design

2.3.1 Evaluation tools and implementation ($\frac{1}{2}$ page)

2.3.2 Research metrics ($\frac{1}{2}$ page)

2.3.3 Evaluation scenarios / Query set ($\frac{3}{4}$ page)

2.3.4 Comparison baselines ($\frac{1}{2}$ page)

2.3.5 Experimental setup and evaluation procedure ($\frac{3}{4}$ to 1 page)

2.4 Reference modeling (1 to $1\frac{1}{2}$ pages)

3. Results

3.1 Prototype output and inference results (2 pages)

3.2 Quantitative evaluation

3.2.1 Retrieval performance ($1\frac{1}{2}$ to 2 pages)

3.2.2 Generation quality ($1\frac{1}{2}$ to 2 pages)

3.2.3 Ablation study (*optional*) - (1 to $1\frac{1}{2}$ pages)

3.3 Qualitative evaluation ($1\frac{1}{2}$ to 2 pages)

3.4 Derived design principles and reference model ($1\frac{1}{2}$ to 2 pages)

4. Discussion

4.1 Main findings (1 page)

4.2 Implications ($\frac{1}{2}$ page)

4.3 Limitations ($\frac{1}{2}$ to 1 page)

4.4 Further research ($\frac{1}{2}$ to $\frac{3}{4}$ page)

Bibliography

Table of figures

List of tables

Table of code

List of abbreviations

Documentation table AI-based tools

Appendices

A1 Literature research

A2 Prototype code

A3 Experiment results

A4 (*placeholder – eventually further artefacts*)

**According to the current outline, the main part of the thesis (from 'Introduction' to the end of 'Discussion') is expected to cover approximately 25 pages.*

References

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